

FTM iMCQ 07

Fall 2025

Genetics, Pharmacology, Histology

FTM2 iMCQ7_KEY

GNET 3 questions L32

A 63-year-old man with prostate cancer is enrolled in a research study that is investigating the role of microRNA in the regulation of cellular proliferation. This type of RNA functions in the cytoplasm and is involved in the regulation of approximately half of all human genes. One specific microRNA is found that is expressed in the patient's cancer cells but is not expressed in normal tissue. Which of the following best explains how this type of RNA alters translation in this tumor tissue?

- | | |
|---------------------------------|-----|
| A. Blocking RNA polymerase | 17% |
| B. Binding to target mRNAs | 17% |
| C. Inhibiting Myc expression | 17% |
| D. Altering mRNA splicing | 17% |
| E. Stabilizing RNA polymerase | 17% |
| F. Increasing half-life of mRNA | 17% |

A 63-year-old man with prostate cancer is enrolled in a research study that is investigating the role of microRNA in the regulation of cellular proliferation. This type of RNA functions in the cytoplasm and is involved in the regulation of approximately half of all human genes. One specific microRNA is found that is expressed in the patient's cancer cells but is not expressed in normal tissue. Which of the following best explains how this type of RNA alters translation in this tumor tissue?

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A 62-year-old woman comes to the physician because of a 7-month history of progressive fatigue and unintentional weight loss. Laboratory studies show anemia and leukocytosis. Genetic tests show upregulation of the *MYC* gene. Which of the following best explains how gene amplification contributes to the clinical findings in this patient?

- | | |
|------------------------------------|-----|
| A. Promotes cell differentiation | 20% |
| B. Promotes cell proliferation | 20% |
| C. Promotes Max/Max dimerization | 20% |
| D. Decreases expression of cyclins | 20% |
| E. Inhibits cell proliferation | 20% |

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An investigator is confirming the work of Jacob and Monod by studying the regulation of the *lac operon* in *Escherichia coli*. Under normal physiological conditions, which of the following best describes the activity of this operon when glucose is abundant and lactose is absent?

- | | |
|---|----|
| A. The 3 structural genes are being expressed >1000x basal levels | 0% |
| B. The 3 structural genes are being expressed at basal levels | 0% |
| C. The repressor protein is bound to the operator | 0% |
| D. cAMP levels in the bacteria are high | 0% |
| E. The CAP protein expression is upregulated | 0% |

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Pharmacology

A 56-year-old man comes to the emergency department because of a 3-day history of fever and pain and swelling to the left leg. He has a 10-year history of diabetes mellitus. He is administered a continuous intravenous infusion of an antibiotic. After careful monitoring, it is determined that the plasma concentration of the antibiotic has reached steady-state despite the lack in improvement of the patient's condition. The physician decides to increase the plasma antibiotic concentration by 100%. Which of the following adjustments to the infusion regimen is the most appropriate to achieve this goal?

- A. Doubling the infusion rate
- B. Halving the infusion rate
- C. Doubling the infusion rate and doubling loading dose
- D. Halving the loading dose and maintaining infusion rate
- E. Doubling the loading dose and maintaining infusion rate

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- B. Halving the infusion rate
- C. Doubling the infusion rate and doubling loading dose
- D. Halving the loading dose and maintaining infusion rate
- E. Doubling the loading dose and maintaining infusion rate

Explanation

$$C_{ss} = \frac{\text{Infusion rate}}{CL}$$

- CL is constant for drugs showing first-order kinetics.
- Therefore the steady-state plasma concentration (C_{ss}), is directly proportional to the infusion rate.
- If infusion rate is doubled, steady-state plasma concentration is doubled.

A 69-year-old man is brought to the emergency department because of a 2-day history of fever and chills. Laboratory studies show MRSA on blood culture. He is to be administered a 200-mg dose of an antibiotic drug intravenously every 12 hours. The drug follows first-order kinetics of elimination with a half-life of 6 hours and a volume of distribution of 20 L. Which of the following is most likely the plasma concentration of the antibiotic immediately before the administration of the second dose?

- A. 10 mg/L
- B. 5 mg/L
- C. 2.5 mg/L
- D. 2 mg/L
- E. 1 mg/L
- F. 0.5 mg/L

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Explanation

Dose = 200 mg

Vd = 20 L

$$\text{Plasma concentration after 1}^{\text{st}} \text{ dose} = \frac{200 \text{ mg}}{20 \text{ L}} = 10 \text{ mg/L}$$

Half-life = 6 hours

Dosing interval = 12 hours = 2 half-lives

10 mg/L

1 half life

5mg/L

2 half lives

2.5 mg/L

A 16-year-old boy is brought to the emergency department because of a 2-week history of fever and cough. Laboratory studies show a positive Group A Streptococcus antigen test. He is to be started on a constant intravenous infusion of an antibiotic at a rate of 2 mg/min to achieve a steady state target concentration of 12 mg/L. The antibiotic follows first-order kinetics of elimination with a volume of distribution of 20 L, and a clearance of 10 L/h. The decision is made to administer the patient with an intravenous loading dose of the antibiotic to achieve the target plasma concentration rapidly. Which of the following is most likely the required loading dose?

- A. 10 mg
- B. 20 mg
- C. 50 mg
- D. 100 mg
- E. 120 mg
- F. 240 mg

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- B. 20 mg
- C. 50 mg
- D. 100 mg
- E. 120 mg
- F. 240 mg

Explanation

$$\text{Loading dose} = V_d \times \text{TC}$$

$$V_d = 20 \text{ L}$$

$$\text{TC} = 12 \text{ mg/L}$$

$$\text{Loading dose} = 20 \text{ L} \times 12 \text{ mg/L} = 240 \text{ mg}$$

A group of investigators is conducting preclinical and clinical trials on five new asthmatic drugs. Each drug has a unique pharmacokinetic and pharmacodynamic profile, as shown in the table below. It is determined that each drug requires a loading dose to rapidly achieve therapeutic plasma concentrations when administered by intravenous injections. Which of the following drugs will most likely require the largest loading dose to achieve the desired plasma concentration?

	Drug	Half-life ($t_{1/2}$)	Volume of Distribution (V_d)
A.	Drug A	4 h	10 L
B.	Drug B	6 h	18 L
C.	Drug C	2 h	66 L
D.	Drug D	15 h	80 L
E.	Drug E	3 h	35 L

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C.	Drug C	2 h	66 L
D.	Drug D	15 h	80 L
E.	Drug E	3 h	35 L

A 45-year-old man comes to the physician for a routine examination. He has a 10-year history of hypertension and kidney failure. His temperature is 37.1°C (98.8°F), pulse is 85/min, blood pressure is 180/110 mm Hg and respirations are 16/min. He is administered an intravenous infusion at a dosing rate of 22 µg/h. The drug is known to have the following pharmacokinetic parameters, a volume of distribution (Vd) of 80 L and a clearance (CL) of 11 L/h. Which of the following is the best approximation of the half-life of the drug in this patient?

- A. 1 h
- B. 5 h
- C. 8 h
- D. 10 h
- E. 15h

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Explanation

$$V_d = 80 \text{ L}$$

$$CL = 11 \text{ L/h}$$

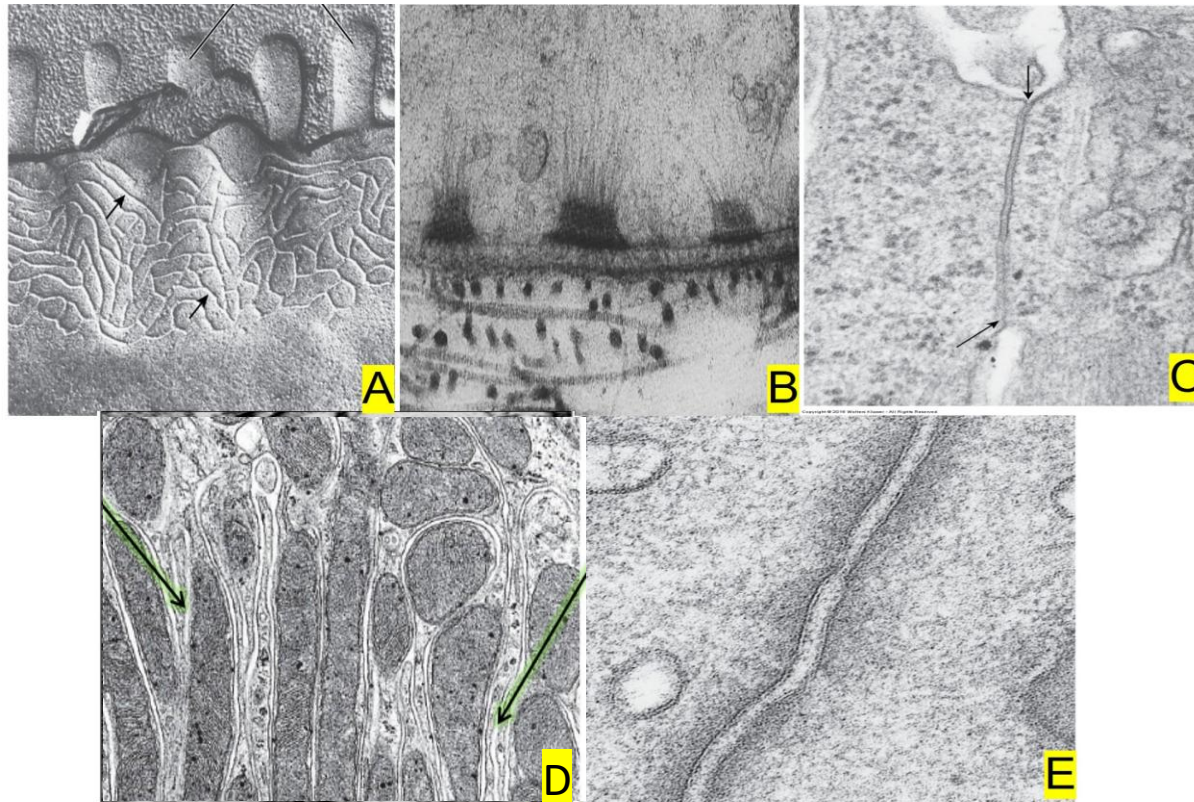
$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

$$t_{1/2} = \frac{0.693 \times 80 \text{ L}}{11 \text{ L/h}} = \frac{55.44}{11} = 5 \text{ h}$$

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Histology

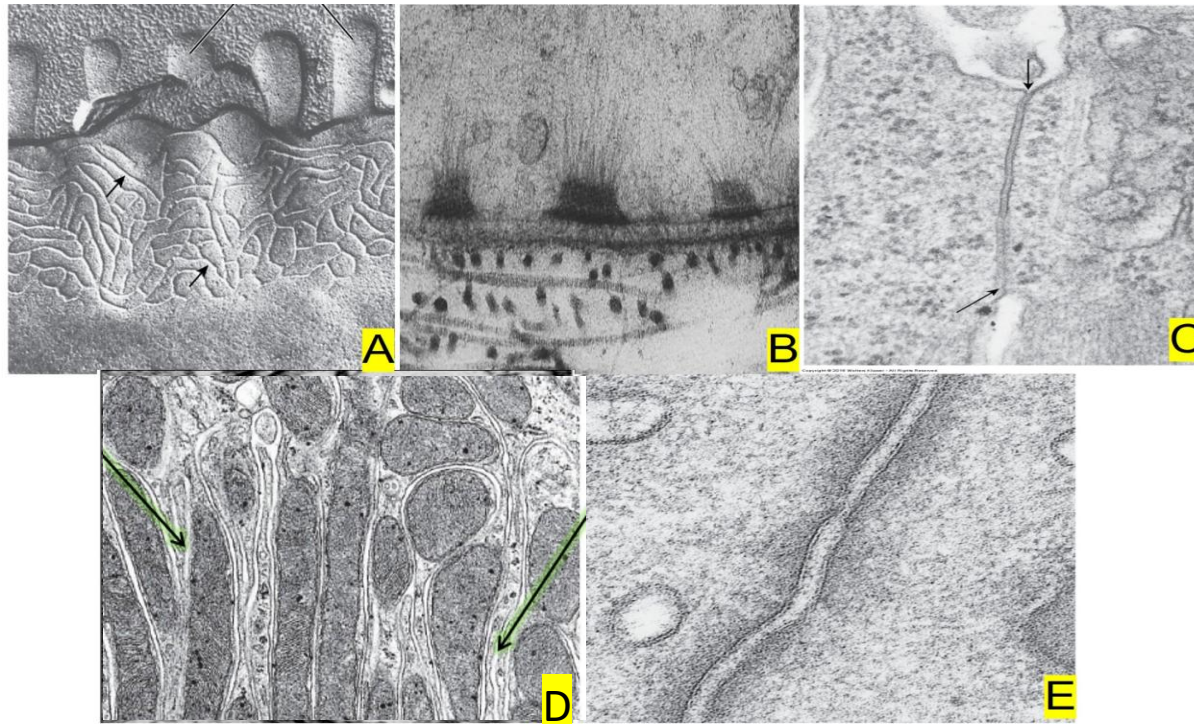
1. A 65-year-old man is brought to the physician because of a 1-hour history of chest pain and shortness of breath, and a 3-month history of intermittent nausea, occasional wheezing and unintentional weight loss of 6.35 kgs (14-lb). His blood pressure is 112/70 mm Hg, pulse is 52/min, and respirations are 25/min. Physical examination shows a mildly uncomfortable man with shallow, rapid respirations. An x-ray shows radio-opacities scattered throughout both lungs. He dies despite appropriate medical care. Autopsy shows lung and heart tissue involvement. Microscopic examination of biopsy samples confirm sarcoidosis. Serology testing shows anti-intercalated disc antibodies and molecular studies show cardiac myocytes with dysfunctional intercalated discs. Which of the following specializations in the attached micrographs most likely forms a component of the dysfunctional structure?



- A. A
- B. B
- C. C
- D. D
- E. E

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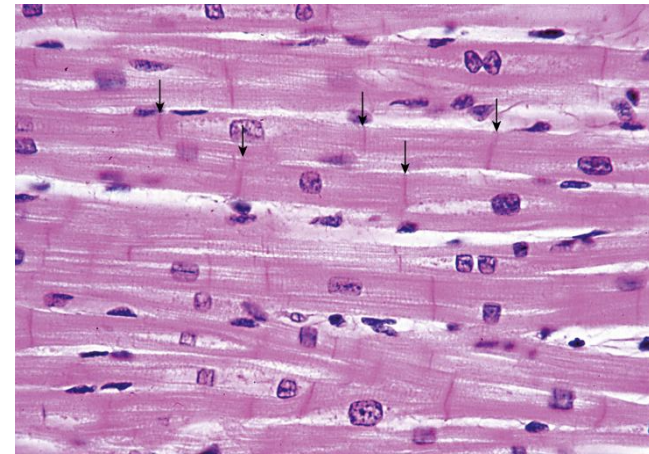
- A. A
- B. B
- C. **C**
- D. D
- E. E



Describe the general organization of cardiac muscle, including the structure and function of intercalated discs. SOM.MK.I.BPM1.1.FTM.3.HCB.0916

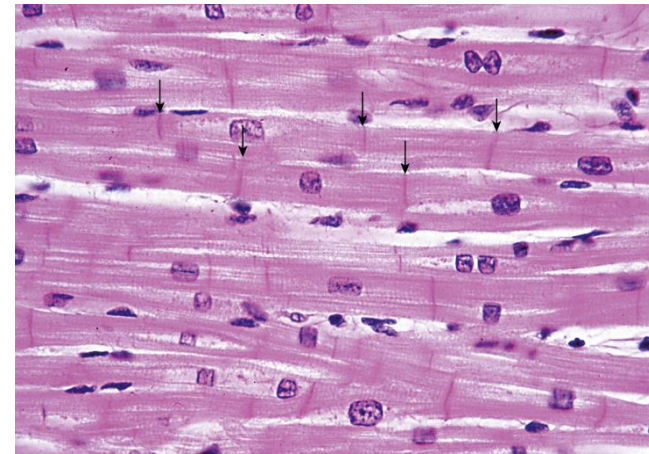
2. A 60-year-old man is brought to the emergency department because of a 2-hour history of intense chest pain that radiates to his left arm, nausea and dizziness. He has a history of type-2 diabetes mellitus and high cholesterol. His blood pressure is 90/50 mm Hg, pulse is 110/min and weak and respirations are 24/min. ECG shows signs of ischemia in leads corresponding to the diaphragmatic surface of the heart. Laboratory studies show elevated troponin T levels. He dies despite appropriately performed life saving measures. Autopsy shows several areas of infarction of the tissue shown in the attached photomicrograph. Which of the following best identifies a function of the elevated protein?

- A. Binds calcium ions in both smooth and striated muscle
- B. Binds calcium ions in striated muscles only
- C. Anchors thin actin filaments at the Z-line
- D. Links actin filaments to the extracellular matrix
- E. Anchors the troponin complex to tropomyosin associated with actin filaments



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3. A 32-year-old woman is brought to the physician because of a 1-hour history of shortness of breath. She has a history of bronchial asthma. Her blood pressure is 101/69 mm Hg, pulse is 112/min, and respirations are 28/min. Physical examination shows severe respiratory distress with cyanosis (blue skin discoloration), increased work of breathing, and decreased air entry in auscultation of the lungs. She is diagnosed with an acute asthma exacerbation. This condition is characterized by hypertrophy of the smooth muscle cells in the walls of the airways called bronchioles. Which of the following features best characterizes these myocytes?

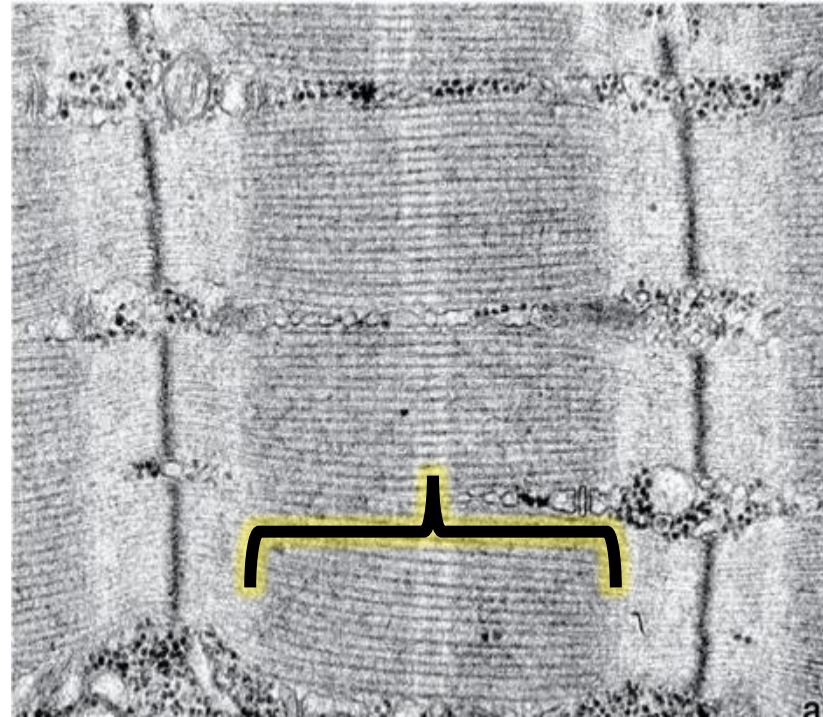
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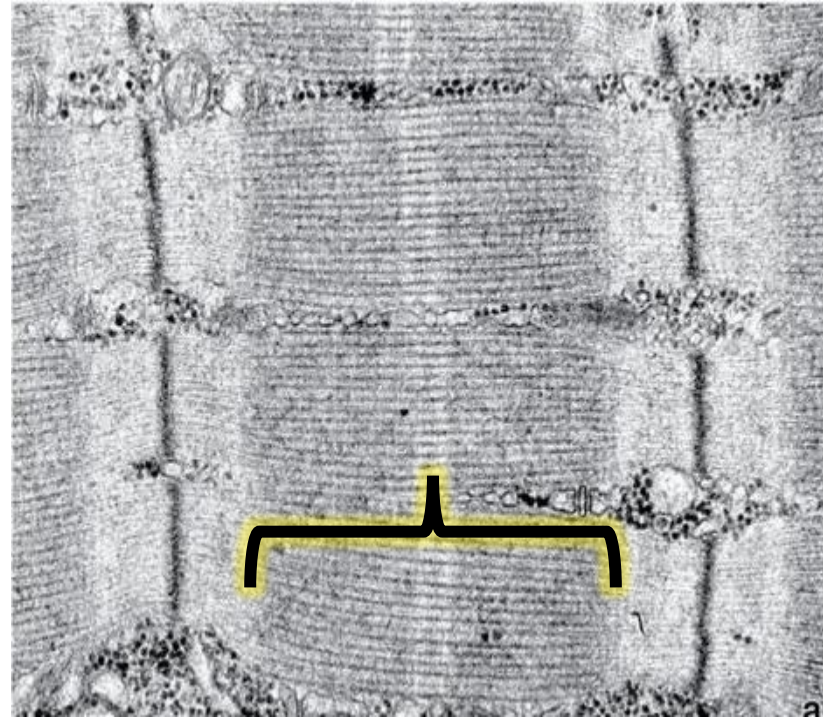
4. A 63-year-old man comes to the physician because of a 1-month history of swelling of his left thigh. Physical examination shows a mass in the quadriceps muscle. Examination of a biopsy specimen of the mass shows poorly differentiated tumor cells that lack the typical cross striations of the normal cells of the tissue examined. A segment of the typical cross striation is outlined within the bracket in the attached TEM image. Which of the following best characterizes this segment?

- A. Contains both thick myosin and thin actin filaments
- B. Has only thick myosin filaments
- C. Represents an area of only thin actin filaments
- D. Localization of the triad in striated muscle
- E. Serves as an anchor for thin actin filaments
- F. Shortens during muscle contraction



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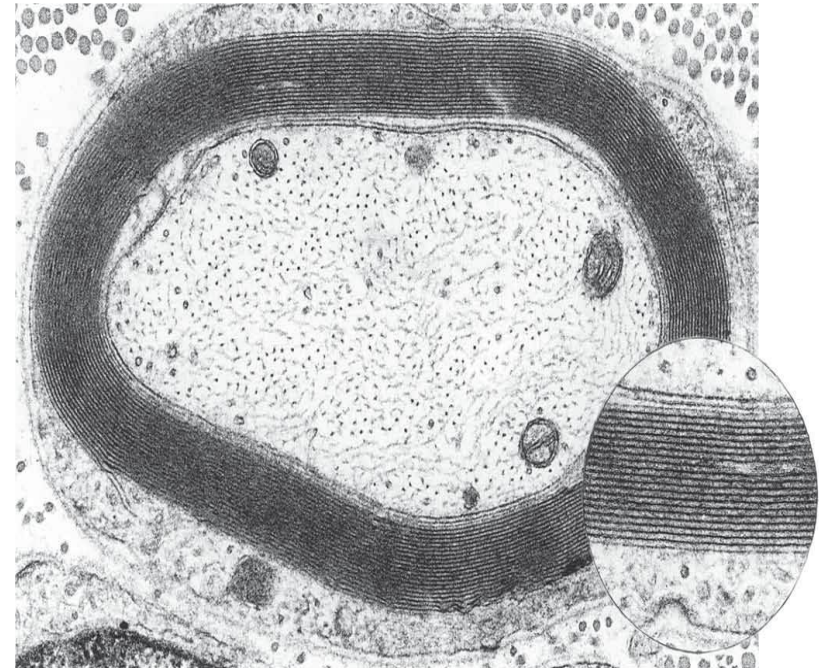


Describe the microscopic structure of the sarcomere.

SOM.MK.I.BPM1.1.FTM.3.HCB.0895

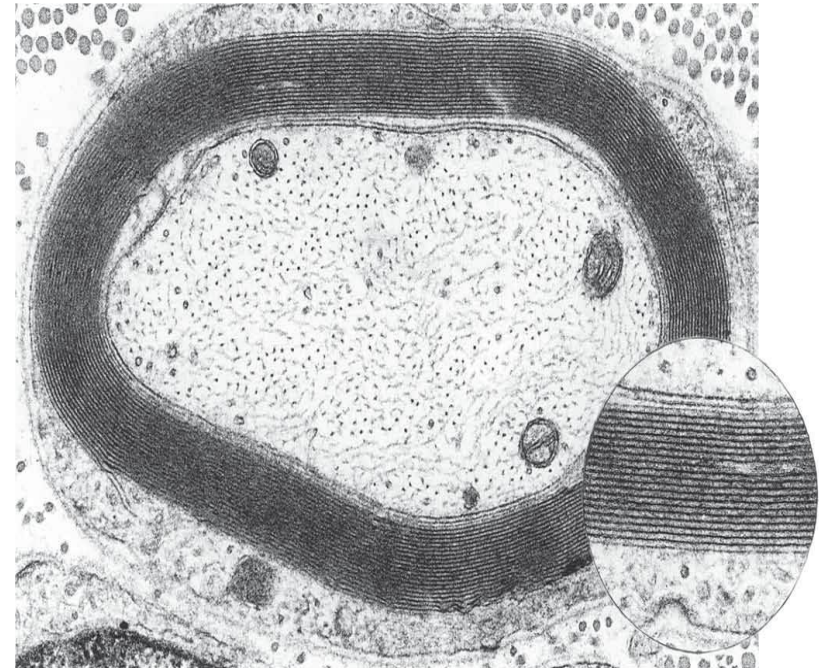
5. A 46-year-old man is brought to the physician because of a 2-week history of weakness in both legs and a 4-day history of inability to walk. He had a viral illness 4-5 weeks ago. Physical examination shows ascending bilateral lower limb muscle weakness (ascending paralysis). A diagnosis of Guillain-Barré syndrome is made. This condition is characterized by damage to the electron dense concentric structure shown in the attached electron micrograph. Which of the following is most likely also a function of the cell type responsible for formation of this structure?

- A. Forms the blood-nerve barrier
- B. Forms the blood-brain barrier
- C. Secretes cerebrospinal fluid
- D. Removal of debris
- E. Regulate neuronal metabolic exchange
- F. Myelination



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List the function(s) of the peripheral neuroglia (Schwann and capsular cells).SOM.MK.I.BPM1.1.FTM.3.HCB.0953

6. A 5-week-old boy is brought to the physician by his mother because of a 1-week history of constipation. Physical examination shows a grossly distended abdomen with palpable dilated loops of colon. Digital rectal examination shows an empty rectal vault and liquid stool with flatus upon withdrawal of the examining finger. Microscopic examination of a biopsy specimen of the distended colon shows an absence of the myenteric plexus in the affected region. A diagnosis of Hirschsprung's disease is made. Which of the following best characterizes the affected nerve tissue?

- A. Presence of pseudo-unipolar or bipolar neurons
- B. Lacks synaptic connections
- C. Alternating cell bodies with axon fascicles
- D. Located in prevertebral and paravertebral regions
- E. Presence of afferent and efferent nerve fibers

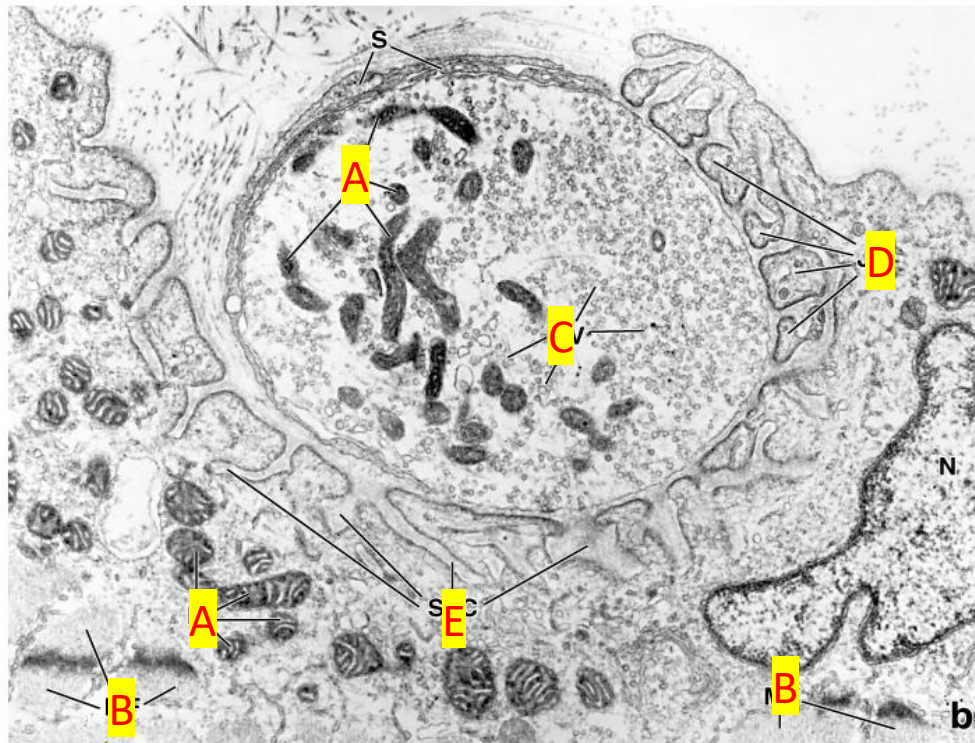
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Describe the microscopic structure of the parasympathetic autonomic ganglia using the myenteric plexus as an example.SOM.MK.I.BPM1.1.FTM.3.HCB.0950

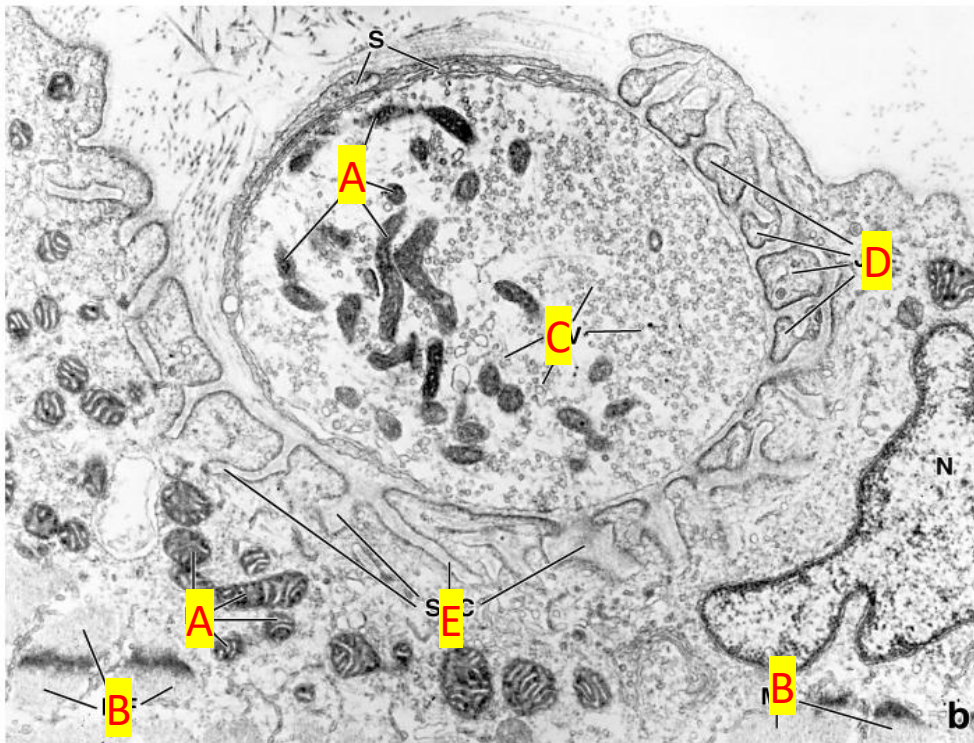
7. A research group is investigating the effect of a new drug for treatment of muscle spasticity. The drug is shown to reversibly bind to acetylcholine receptors in the skeletal muscle cells of experimental rodents. Which of the following is the most likely site of action of the drug?

- A. A
- B. B
- C. C
- D. D
- E. E



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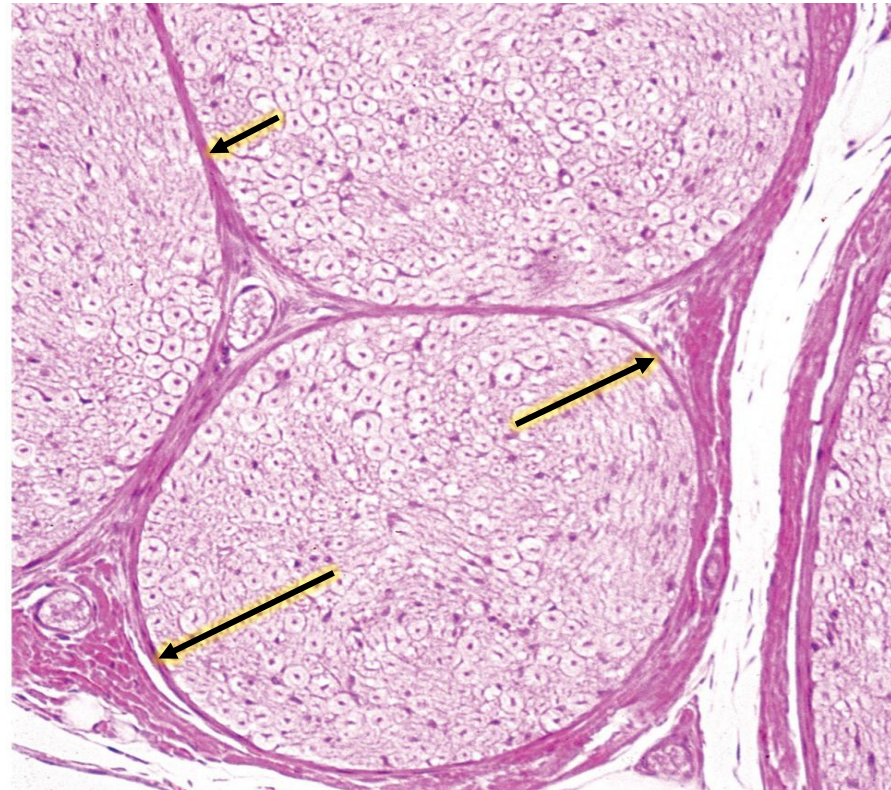
- A. A
- B. B
- C. C
- D. D
- E. E



Describe the histological structure of the neuromuscular junction.
SOM.MK.I.BPM1.1.FTM.3.HCB.0939

8. A 25-year-old woman comes to the physician because of a 2-week history of numbness to the dorsal surface of the right hand. She has a history of removal of a cystic mass to that site 6 weeks ago. Physical examination shows loss of sensation on the dorsum of the right hand. Examination of a biopsy specimen of the dorsum of the hand shows inflammation of the tissue shown at the tips of the arrows in the attached photomicrograph. Which of the following structures is likely affected?

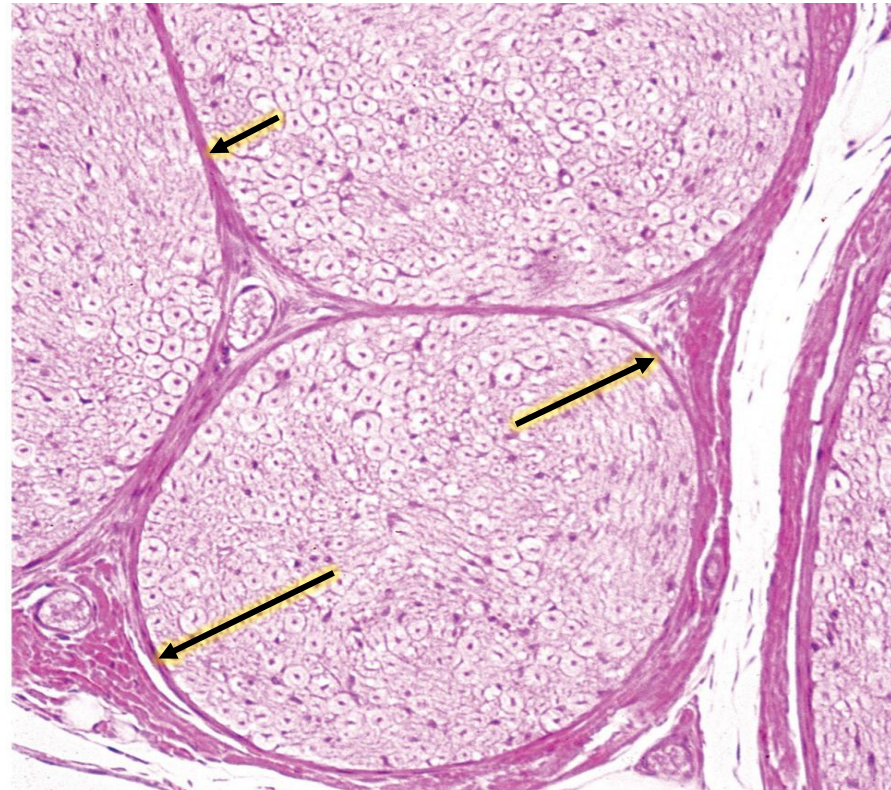
- A. Endomysium
- B. Endoneurium
- C. Epimysium
- D. Epineurium
- E. Myelin sheath
- F. Perimysium
- G. Perineurium



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- E. Myelin sheath
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- G. Perineurium



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Describe the connective tissue sheaths of a peripheral nerve.SOM.MK.I.BPM1.1.FTM.3.HCB.0958